

Tejdeep Pathipati

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EDUCATION

George Mason University

Fairfax, Virginia

Bachelor of Science in Computer Science, GPA: 3.65

Expected Graduation: May 2026

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (Java), Low-Level Programming (C), Databases, Operating Systems, Software Engineering, Data Mining, AI

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (MySQL), JavaScript

Frameworks: React, Node.js, Flask, JUnit, RestAPI, FastAPI

Cloud & DevOps: AWS (S3, EC2, EMR, Glue, Lambda, SageMaker), Docker, Azure, Kubernetes, Git

AI/ML Frameworks: TensorFlow, PyTorch, Scikit-Learn, Pandas, NumPy

Awards & Certifications: Drone Prototyping, AWS AI Practitioner, Scaled Agile Framework

EXPERIENCE

Data Engineering Intern

May 2025 - Aug 2025

Wi-Tronix LLC

Bolingbrook, Illinois

- Developed a data management pipeline tool to filter and curate 50,000+ daily images and metadata, streamlining access for analytics and ML model training.
- Architected a cloud-native data lakehouse on AWS S3, EKS, Iceberg Tables, Athena, automating ingestion of 5TB+. Managed metadata with PyIceberg and SQL to enable reproducible versioning, cut manual ETL 50%, and delivered QuickSight dashboards.
- Worked in a Agile (SAFe) team while integrating ML models with Comet ML and processing lab audio, expanding voice-to-text training data by 25%.

Undergraduate Teaching Assistant - Python

Aug 2024 - May 2025

George Mason University, College of Engineering and Computing

Fairfax, Virginia

- Led weekly lab sessions for 50+ students, focusing on data structures, algorithm design, and object-oriented principles, while developing practical coding exercises to reinforce key concepts.
- Assisted 70+ students with debugging and improving code quality, emphasizing software development practices.
- Provided post-project feedback, optimizing code efficiency and improving student grades by 10%.

Software Engineer Intern

May 2024 - Aug 2024

Rise Consultancy Edu

Remote, India

- Built a scalable backend system in Python and MySQL, optimizing data storage and retrieval for 200+ active users
- Designed, deployed, and secured RESTful APIs, integrating authentication (OAuth, JWT) and rate limiting, reducing API response times by 35% through query optimization and indexing.
- Implemented a caching layer (Redis) to minimize redundant database queries, reducing system load by 30% and enhancing application performance for high-concurrency requests.

PROJECTS

Custom Transformer LLM | *Python, PyTorch, Transformers, "Attention Is All You Need"* - [GitHub](#)

- Implemented a custom Transformer model with encoder and decoder based on "Attention Is All You Need" paper.
- Designed an end-to-end training pipeline with label smoothing, early stopping, and beam search decoding.
- Evaluated model performance using BLEU, ROUGE-L, and token-level F1 scores on the DailyDialog dataset.

YouTube Data Analysis ELT Pipeline | *Python, AWS (S3, Glue, Lambda, Athena), QuickSight* - [GitHub](#)

- Developed a scalable ELT pipeline on AWS to automate processing of 10K+ YouTube trending records per day.
- Transformed JSON data into optimized Parquet format using Lambda functions for efficient querying in Athena.
- Visualized regional viewing trends and engagement metrics across 15+ countries via QuickSight dashboards

Chest Cancer Classification | *Python, TensorFlow, MLflow, DVC, AWS* - [GitHub](#)

- Built an end-to-end CNN pipeline to classify breast cancer from medical imaging with 85% accuracy.
- Tracked 5+ model experiments with MLflow, enabling reproducible results and easier performance comparison.
- Optimized data handling with DVC and AWS S3, reducing processing time by 20%.